CNNs Under the Hood: Understanding Feature Learning

Monday, 2 November 2026 10:48 PM

Intuition.

Image → 32 × 32 × 3 (h × w × c)

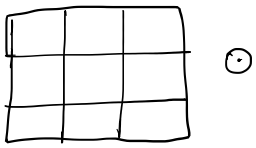
padding = same, Conv1 = 3 × 3 × 3, # of kernels / filters = 16

↓

32 × 32 × 16

Padding = 0. valid.

No padding.



2 × 2.

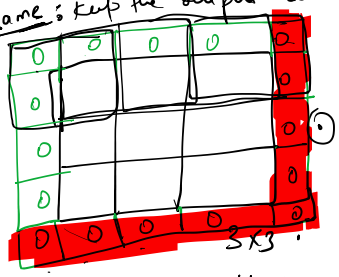
stride = 1, padding = 0

$$\frac{3 - 2 + 2 \times 0}{1} + 1$$

3 × 3

Padding = same : keep the output dimension same as input. 2 × 2.

adds 1R, 1C



stride = 1

$$\frac{3 - 2 + 2 \times 1}{1} + 1$$

2 × 2.

$$\frac{4 - 2 + 2 \times 0}{2} + 1 = 3.$$

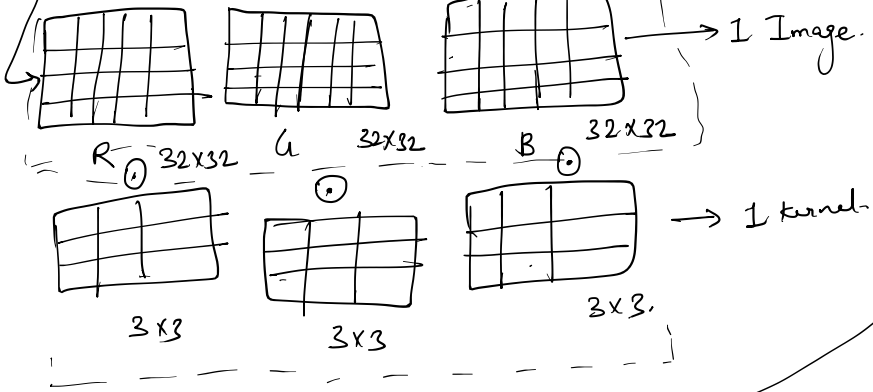
3 × 3.

4 × 4.

Step 1:

Image → 32 × 32 × 3

padding = same, Conv1 = 3 × 3 × 3 # of conv = 16.



× 16

Step 2.

32 × 32 × 16.

h w c

↓ Maxpool — 2 × 2 stride = 2.

Step 3.

16 × 16 × 16

h w c

padding = same, Conv2 — 3 × 3 × 16 # Conv = 32

↓

16 × 16 × 32

Step 4:

↓ Maxpool – 2×2 , S = 2.

Step 5:

8×8×32

padding = same ↓ Conv 3 – 3×3×32 # of K = 64.

Step 6:

8×8×64 – Maxpool – 2×2 , S = 2.

↓

Step 7:

4×4×64. – Conv 4 – 3×3×64, # of K = 10.
padding = same.

↓

Step 8:

4×4×10

↓ GAP.

Step 9:

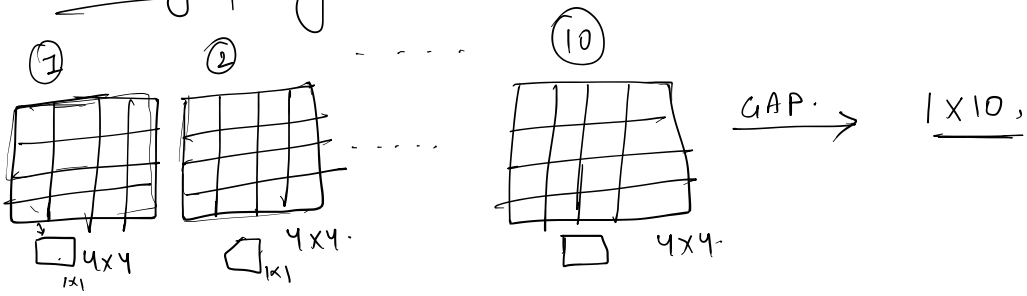
1×1×10

↓ Softmax Layer.

Step 10:

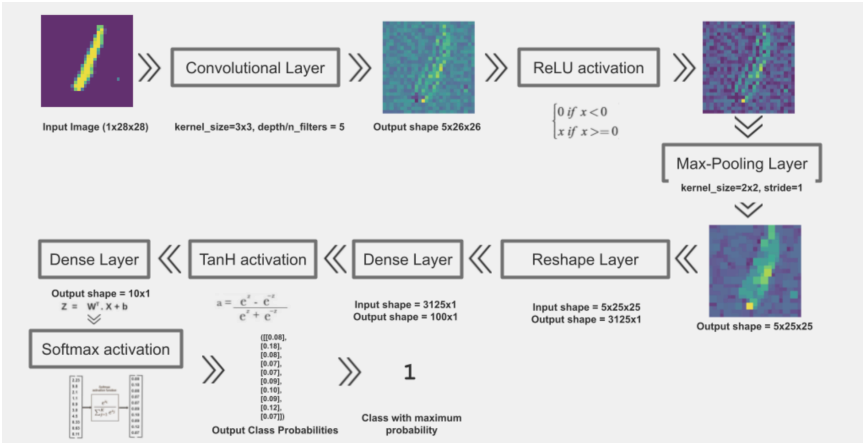
10 probabilities (10 classes). (0–9):

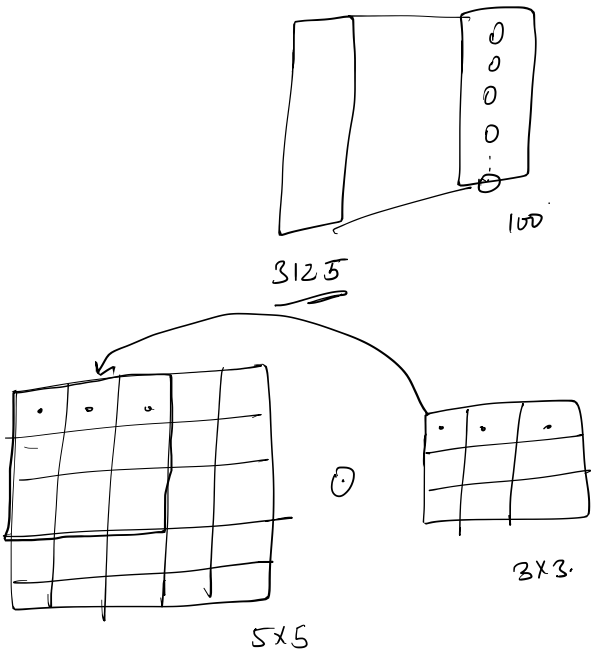
GAP: Global avg pooling



4×4×10 → 1×10.

FC → NN.

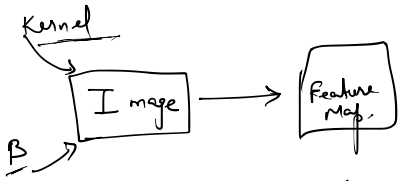




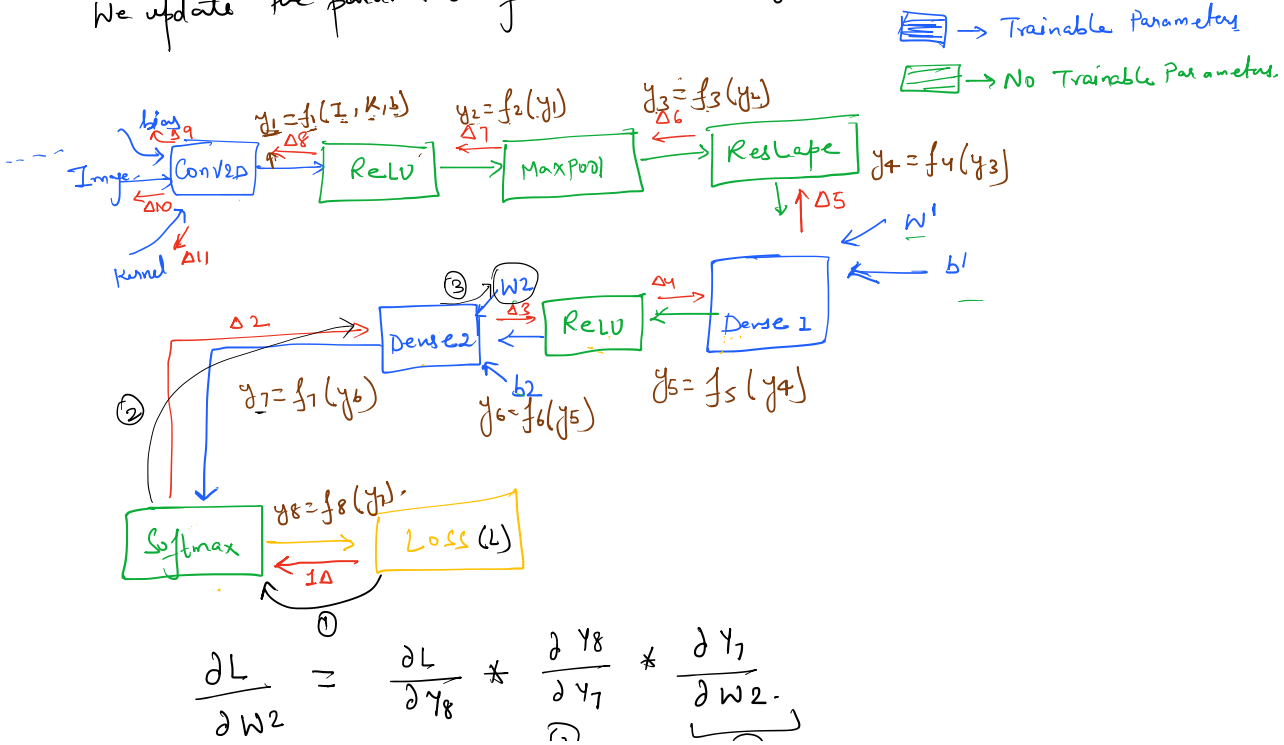
✱ Backward prop

Initially the kernel and bias also will have random values.

They will extract random features.

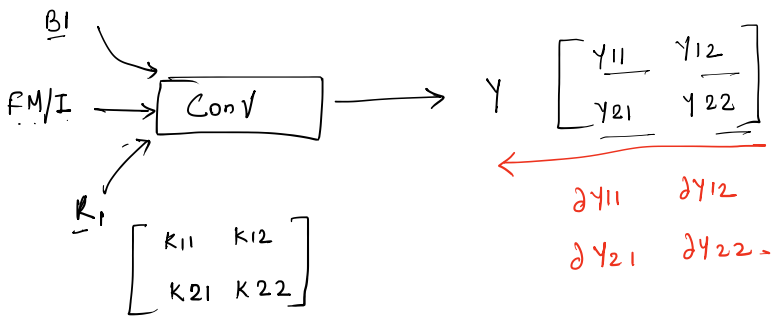


We update the parameters of our model using backprop.



$$\frac{\partial L}{\partial W2} = \frac{\partial L}{\partial y_8} * \frac{\partial y_8}{\partial y_7} * \frac{\partial y_7}{\partial W2}$$

① ② ③



$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \odot K_1 \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} + B_1$$

$$y_{11} = x_{11} k_{11} + x_{12} k_{12} + x_{21} k_{21} + x_{22} k_{22} + b.$$

$$y_{12} = x_{12} k_{11} + x_{13} k_{12} + x_{22} k_{11} + x_{23} k_{22} + b.$$

$$y_{21} = x_{21} k_{11} + x_{22} k_{12} + x_{31} k_{21} + x_{32} k_{22} + b.$$

$$y_{22} = x_{22} k_{11} + x_{23} k_{12} + x_{32} k_{21} + x_{33} k_{22} + b.$$

$$\frac{\partial L}{\partial K_1} = \begin{bmatrix} \frac{\partial L}{\partial k_{11}} & \frac{\partial L}{\partial k_{12}} \\ \frac{\partial L}{\partial k_{21}} & \frac{\partial L}{\partial k_{22}} \end{bmatrix}$$

$$\frac{\partial L}{\partial k_{11}} = \frac{\partial L}{\partial y_{11}} \cdot \frac{\partial y_{11}}{\partial k_{11}} + \frac{\partial L}{\partial y_{12}} \cdot \frac{\partial y_{12}}{\partial k_{11}} + \frac{\partial L}{\partial y_{21}} \cdot \frac{\partial y_{21}}{\partial k_{11}} + \frac{\partial L}{\partial y_{22}} \cdot \frac{\partial y_{22}}{\partial k_{11}}$$

x_{11}, x_{12}, x_{21}, x_{22}

$$\frac{\partial L}{\partial K_1} = \begin{bmatrix} \frac{\partial L}{\partial k_{11}} & \frac{\partial L}{\partial k_{12}} \\ \frac{\partial L}{\partial k_{21}} & \frac{\partial L}{\partial k_{22}} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} \Theta \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{12}} \\ \frac{\partial L}{\partial y_{21}} & \frac{\partial L}{\partial y_{22}} \end{bmatrix}$$

$$k_1 - \alpha \frac{\partial L}{\partial k_1}$$

LR.

Bias.

$$\frac{\partial L}{\partial B} = \frac{\partial L}{\partial Y} \cdot \frac{\partial Y}{\partial B}$$

$$\frac{\partial L}{\partial B} = \frac{\partial L}{\partial Y_{11}} \cdot \frac{\partial Y_{11}}{\partial B} + \frac{\partial L}{\partial Y_{12}} \cdot \frac{\partial Y_{12}}{\partial B} + \frac{\partial L}{\partial Y_{21}} \cdot \frac{\partial Y_{21}}{\partial B} + \frac{\partial L}{\partial Y_{22}} \cdot \frac{\partial Y_{22}}{\partial B}$$

$$\frac{\partial L}{\partial B} = \text{sum} \begin{bmatrix} \partial L/\partial Y_{11} & \partial L/\partial Y_{12} \\ \partial L/\partial Y_{21} & \partial L/\partial Y_{22} \end{bmatrix}$$

How can we calculate the gradients to update Kernel weights in Convolutional Layer during Backpropagation ?

B) Calculating derivative of loss with respect to kernel weights

Quiz 3.

weights = [-3, 2.8, -1]
input = [4, 2, 2]
Bias = 5

((-3).(4) + (2.8).(2) + (-1).(2)) + 0.5
= (-12 + 5.6 - 2) + 0.5
= -8.4 + 0.5
Y → = -7.9

$$ReLU(W \cdot X + b)$$

$$ReLU \rightarrow \max(Y, 0)$$